Mongolia International Mathematics Competition 2026

Ulaanbaatar City, 22nd to 27th July 2026

Elementary Mathematics International Contest

Team Contest

Time limit: 70 minutes

Information:

- You are allowed 70 minutes for this paper, consisting of 10 questions printed on separate sheets. For questions 1, 3, 5, 7 and 9, only numerical answers are required. For questions 2, 4, 6, 8 and 10, full solutions are required.
- Each question is worth 40 points. For odd-numbered questions, no partial credits are given. There are no penalties for incorrect answers, but you must not give more than the number of answers being asked for. For questions asking for several answers, full credit will only be given if all correct answers are found. For even-numbered questions, partial credits may be awarded.
- Diagrams shown may not be drawn to scale.

Instructions:

- Write down your team's name and team number in the space provided on every question sheet.
- Enter your answers at the bottom right corners of the corresponding question sheets.
- During the first 10 minutes, the four team members examine the first 8 questions together and discuss them. After 10 minutes they distribute the questions among themselves, with every team member allotted at least 1 question.
- During the next 35 minutes, the four team members write down the solutions of their allotted problems on the respective question sheets, with no further communication or discussion.
- During the last 25 minutes, the four team members work together to write down the solutions of the last 2 questions on the respective question sheets.
- It is forbidden to use instruments such as protractors, calculators and electronic devices, smart watches.
- At the end of the contest, you must hand in the envelope containing all question sheets and all scrap paper.

English Version

Team: _____ Team No.: _____

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1. As shown in Figure 1, there are three different slopes which are formed from three mutually tangent circular arcs: a quarter-circle of radius 21 cm, followed by a quarter-circle of radius 35 cm, followed by a semicircle of radius 21 cm. As shown in Figure 2, a disk of radius 7 cm rolls along the slope from the left endpoint to the right endpoint, without slipping and always remaining tangent to the slope. What is the total area, in cm^2, of the region covered by the disk as it rolls along the slope? (Note: Take $\pi = \frac{22}{7}$)

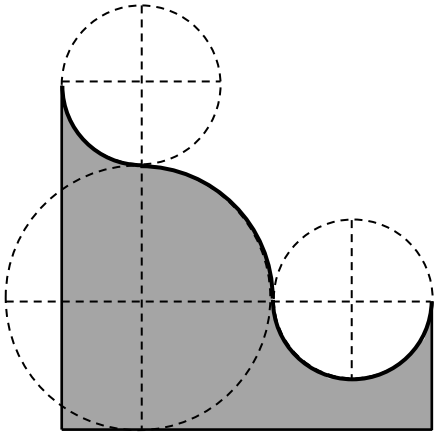


Figure 1

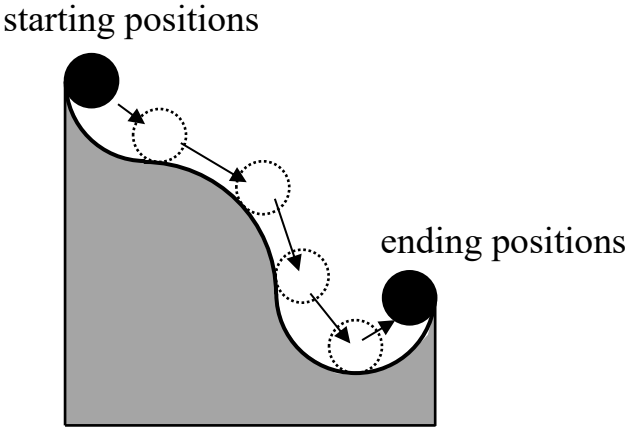


Figure 2

Answer: _____ cm^2



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2. What is the smallest prime number p, such that p^3+3p^2 has exactly 18 positive divisors?

Answer: _____

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3. Alex begins by writing down the number 100 on a piece of paper. Starting from the second number onward, each number written on the paper is the sum of the previous number and its largest prime factor. For example, the second number is $100+5=105$ since the largest prime factor of 100 is 5, and the third number is $105+7=112$ since the largest prime factor of 105 is 7. What is the smallest possible number written by Alex that is greater than or equal to 2026?

Answer: _____

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4. A palindrome is a word that is spelled the same forwards and backwards. When using the letters from the word NAADAM, nine different palindromes can be formed, which are:

A, D, M, N, AA, AAA, ADA, AMA, ANA.

How many different palindromes can be formed by using the 11 letters of MONGOLIA IMC ?

Answer: _____ different palindromes

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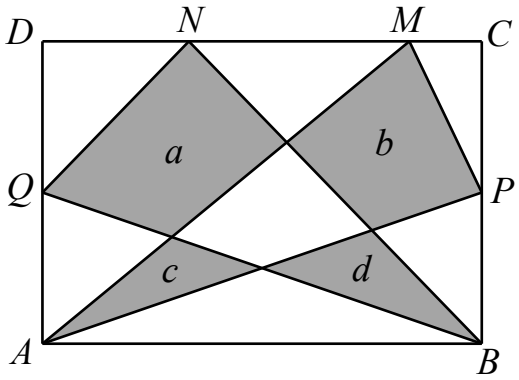
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5. As shown in the diagram below, the area of rectangle $ABCD$ is 408 cm^2. Points M and N lie on side CD, such that $DN : NM : MC = 2 : 3 : 1$. Points P and Q are the midpoints of sides BC and AD, respectively. If the areas of the four shaded regions are a, b, c and d cm^2, respectively, what is the value of $(a+d)-(b+c)$?



Answer: _____

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6. Let A, B, C and D be four distinct positive integers such that

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}+\frac{1}{D}=\frac{21}{22}.$$

What are all possible values of $A+B+C+D$?

Answer: _____

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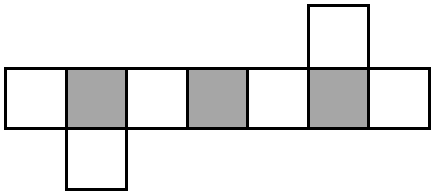
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7. Place the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 into each of the nine squares shown in the diagram below, using each digit exactly once.



The digit in each shaded square (▩) must be at least 3 greater than the digit in every square sharing a side with it.

For example, if a shaded square contains the digit 9, all squares sharing a side with it may contain any of the digits 1, 2, 3, 4, 5 or 6 but not 7 or 8.

In how many ways can the digits be placed in the diagram?

Answer: _____ ways

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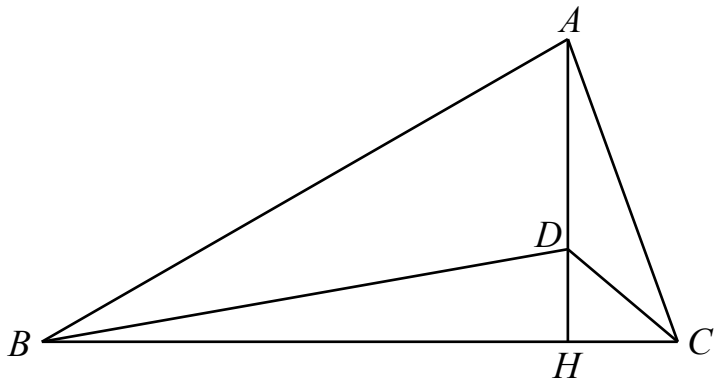
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8. As shown in the diagram below, triangle ABC is acute and AH is the altitude from A to side BC. Point D lies on segment AH, where $\angle DBA = \angle DAC = 20°$ and $\angle DBC = 10°$. What is the measure, in degrees, of $\angle DCB$?



Answer: _____ °

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9. A Mongolian yak herdsman manages two different grazing areas: Pasture A and Pasture B. In each pasture, the grass grows at a constant daily rate, although the growth rates may differ between the two pastures. Each yak consumes grass at the same constant rate.

Through years of experience, the herdsman has made the following observations:

For Pasture A:

- 29 yaks can completely consume all the grass in exactly 7 days,
- 17 yaks can completely consume all the grass in exactly 28 days.

For Pasture B:

- 41 yaks can completely consume all the grass in exactly 6 days,
- 21 yaks can completely consume all the grass in exactly 26 days.

The herdsman currently owns 32 yaks, all of which must graze every day. On any given day, he may divide the yaks between the two pastures in any way he chooses, and the allocation of yaks may vary from day to day.

What is the maximum number of days the 32 yaks can continue grazing before the grass in both pastures is completely consumed?

Answer: _____ days

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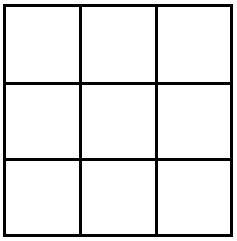
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10. In a 3×3 grid, at least one cell is to be shaded. In how many ways can the grid be shaded such that no two shaded cells have a common side?
(Note: Shaded cells are allowed to meet at corners.)



Answer: _____ ways